A High-Efficiency Push–Pull Parallel-Circuit Class-E/F$_3$ Power Amplifier for Harmonic Suppression

Heng Lu, Jianliang Jiang, and Hengli Zhang

***Abstract*— In this article, a high-efficiency push–pull parallel-circuit (PC) Class-E/F$_3$ power amplifier (PA) with a harmonic suppression network for radio frequency identification (RFID) applications is presented. With the double reactance compensation technique (D-RCT), the PA's design of broadband performance is achieved. In addition, a low-pass (LP) Chebyshev technique is introduced to provide a simple fundamental matching network (MN). Finally, a high-efficiency push–pull PC Class-E/F$_3$ PA is fabricated and measured. The experimental results illustrate that an output power (P_{out}) is from 36.83 to 41.84 dBm and 82.35%–91.32% drain efficiency (DE) operating from 5 to 10.5 MHz frequency range, which agrees well with the simulation results.**

***Index Terms*— Class-E/F$_3$, high-efficiency, parallel-circuit (PC), power amplifier (PA), push–pull.**

I. Introduction

With the rapid development of modern wireless communication technology, the demand for highly efficient broadband power amplifiers (PAs) is urgently needed to meet various requirements. Techniques like switch-mode PAs (SMPAs) [1], [2], [3], [4], [5] ensure that the drain current and voltage waveforms have a minimal overlap in the time domain, leading to nearly 100% theoretical efficiency. At the same time, this technology has been widely used in the RF and microwave industry due to its advantages such as design simplicity, high reliability, and easy integration.

Recently, many hybrid PAs have been reported for improving the PA's performance by employing the *LC* resonance circuit or 1/4-wavelength transmission line. For example, a single-ended Class-EF PA is presented and analyzed in [6], which can provide a peak drain voltage (V_s) of two times the dc voltage (V_{DD}) for driving signals with duty cycles $D < 50\%$. The variant Class-EF$_2$ is reported in [7] to tune the second harmonic, offering a V_s of $3.46V_{DD}$ for a duty cycle of $D = 50\%$. However, an RF choke L_{RFC} is used in these cases. To reduce the circuit complexity. In [8], a parallel-circuit (PC) Class-EF PA with a duty cycle D of 1/3 is investigated, leading to a significant improvement in V_s and f_{max}. However, the PC Class-E/F$_2$ configuration in [9] provides a higher V_s of $3.68V_{DD}$ compared to the Class-EF$_2$. Tuning the third harmonic is a good method to reduce the V_s. In [10], the Class-E/F$_3$ can offer a V_s of 3.14 V_{DD}, and a PC Class-E/F$_3$ offers a V_s of 3.28 V_{DD}, but the f_{max} is 1.38 times that of Class-E/F$_3$ [11]. Similar configurations and analysis can also be found in [12], which greatly enhances their applicability in practical circuit implementation. In [13], we proposed a single-ended PC Class-E PA in which higher harmonics are only filtered out by using a low-pass (LP) matching network (MN). However, the wide bandwidth and good harmonic suppression are difficult to achieve simultaneously.

In this study, a push–pull PC Class-E/F$_3$ PA is introduced. By utilizing a double reactance compensation technique (D-RCT) [14], the load network is analyzed and derived to broaden the frequency band of the PA. The third harmonic is tuned by using series resonator L_3–C_3, while the elimination of higher even harmonics can be achieved through the push–pull configuration, thereby reducing not only substantial energy losses but also EMI issues of the PA. As a result, the proposed push–pull PC Class-E/F$_3$ PA shows promise for applications in RFID and IoT systems.

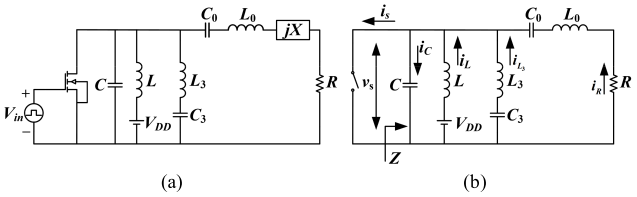


Fig. 1. (a) Basic circuit of the PC Class-E/F$_3$. (b) Equivalent circuit of PC Class-E/F$_3$ with D-RCT in (a).

II. Single-Ended PC Class-E/F$_3$ With D-RCT Analysis

Fig. 1(a) shows the basic circuit of the PC Class-E/F$_3$, and the equivalent circuit of the PC Class-E/F$_3$ with D-RCT is shown in Fig. 1(b). In this case, the series phase-shifting reactance *jX* in Fig. 1(a) is removed and a D-RCT is applied to broaden the bandwidth. The series resonators L_3–C_3 and L_0–C_0 are tuned at $3f_0$ and f_0, respectively, providing a short circuit, while the parallel-tuned circuit L–C tuned to f_0 provides an open circuit. The circuit is analyzed as follows.

For a lossless operation mode, two idealized optimum conditions should be satisfied [15], [16]

$$v_s(\omega t)|_{\omega t=2\pi} = 0 \tag{1}$$

$$dv_s(\omega t)/d\omega t|_{\omega t=2\pi} = 0 \tag{2}$$

where $v_s(\omega t)$ is the switch voltage and ω is the angular frequency. The load network impedance Z can be expressed

Manuscript received 12 June 2024; revised 18 July 2024 and 4 August 2024; accepted 8 August 2024. Date of publication 20 August 2024; date of current version 9 October 2024. This work was supported in part by China Scholarship Council. *(Corresponding author: Jianliang Jiang.)*

The authors are with the School of Optics and Photonics, Beijing Institute of Technology, Beijing 100081, China (e-mail: 15090378052@163.com; jiangjianliang@bit.edu.cn; zhl040325@bit.edu.cn).

Color versions of one or more figures in this letter are available at https://doi.org/10.1109/LMWT.2024.3442206.

Digital Object Identifier 10.1109/LMWT.2024.3442206

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as follows:

$$Z = 1/j\omega C \parallel j\omega L \parallel (R + j\omega L_0 + 1/j\omega C_0) \parallel \times (j\omega L_3 + 1/j\omega C_3). \quad (3)$$

The admittance Y is obtained as follows:

$$Y = 1/Z = j\omega C + 1/j\omega L + 1/(j\omega L_3 + 1/j\omega C_3) + 1/(j\omega L_0 + 1/j\omega C_0 + R). \quad (4)$$

Then the load network susceptance is given by the following equation:

$$B(\omega) = \mathrm{Im}Y = \omega C - 1/\omega L + \omega C_3/(1 - \omega^2 L_3 C_3) + \omega C_0(1 - \omega^2 L_0 C_0)/\left((1 - \omega^2 L_0 C_0)^2 + \omega^2 C_0^2 R^2\right). \quad (5)$$

To obtain a constant phase angle of the impedance by the load network to the transistor drain reference plane in a broadband frequency bandwidth, the L_0–C_0 values should be chosen carefully. Therefore, the D-RCT is applied through finding zero of the first and second derivatives of $B(\omega)$ at the center angular frequency as follows:

$$dB(\omega)/d\omega\big|_{\omega=\omega_0} = d^2B(\omega)/d\omega^2\big|_{\omega=\omega_0} = 0. \quad (6)$$

Then, the equation systems can be simplified as the two following equations:

$$C + 1/L\omega_0^2 - 2/C_0R^2\omega_0^2 + 45C_3/32 = 0 \quad (7)$$

$$-2/L\omega_0^3 + 2/C_0R^2\omega_0^3 + 63C_3/64\omega_0 = 0. \quad (8)$$

Equations (7) and (8) are required to be solved simultaneously, as a result, the series capacitance C_0 and C_3 are obtained by the following equation:

$$C_0 = 34L/\left(R^2(7CL\omega_0^2 + 27)\right) \quad (9)$$

$$C_3 = -64(CL\omega_0^2 - 1)/(153L\omega_0^2). \quad (10)$$

Then, the parameters of the series inductors L_0, L_3 resonate with C_0 at f_0 and resonate with C_3 at $3f_0$, respectively, which can be calculated as follows:

$$L_0 = 1/\left(34\omega_0^2 L/\left(R^2(7CL\omega_0^2 + 27)\right)\right) \quad (11)$$

$$L_3 = 1/\left(-64(3\omega_0)^2(CL\omega_0^2 - 1)/(153L\omega_0^2)\right). \quad (12)$$

For the specified supply voltage of V_{DD}, the output power of P_{out}, and center operating angular frequency of ω_0, the optimum values of the load resistance R, parallel inductance L, and capacitance C can be determined using the following equations [11]:

$$R = 1.256V_{DD}^2/P_{out} \quad (13)$$

$$L = 0.975R/\omega_0 \quad (14)$$

$$C = 0.551/\omega_0 R \quad (15)$$

and the phase angle seen from the drain node is represented as follows:

$$\tan\phi = R/\omega L - \omega RC. \quad (16)$$

Substituting (14) and (15) into (16), then the phase angle at fundamental is 25.39°. For V_{DD} = 12 V and P_{out} = 5 W, the optimum R obtained by (13) is 36.2 Ω, as shown by the dashed line in Fig. 2(a). The real and imaginary parts of the load network impedance versus frequency calculated by (3) are shown in Fig. 2(a), indicating the proposed PC Class-E/F$_3$ with D-RCT can provide constant resistance and reactance over a wide frequency range.

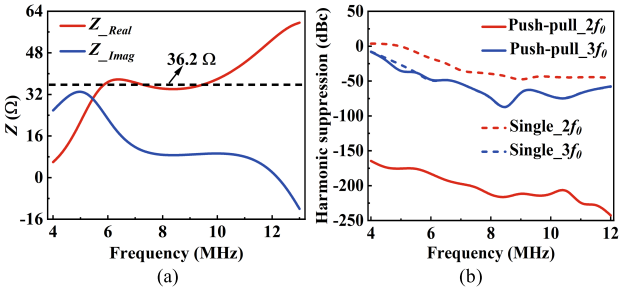


Fig. 2. (a) Real and imaginary parts of the load network impedance versus frequency. (b) $2f_0$ and $3f_0$ harmonic suppression levels versus frequency.

III. Push–Pull PC Class-E/F$_3$ Circuit Design

The push–pull configuration is implemented to work symmetrically, which consists of two individual active devices and operates as switches to drive periodically. The gate drive signals are amplitude identical with a 180° phase shift. Then, the load voltages are differentially summed through a balun, eliminating even harmonics at the load termination. To validate the analysis of single-ended and push–pull structures, the simulation is performed using the advanced designed system (ADS) platform and based on harmonic balance analysis in order of 5. The simulated $2f_0$ and $3f_0$ harmonic suppression levels of the single and push–pull structures are depicted in Fig. 2(b). The proposed PA displays superior $2f_0$ and $3f_0$ harmonic suppression over approximately 50 dBc in the range of 6–12 MHz. The $2f_0$ harmonic suppression is obtained due to the elimination of even harmonic contents through a push–pull configuration. The $3f_0$ harmonic suppression is obtained because the series resonant circuit L_3–C_3 presents very low or zero impedance at $3f_0$. A second-order LP-MN is employed to convert the optimal impedance to the 50 Ω load, and then a 2:1 (impedance ratio) balun is used as a transformation block between the differential termination and a single-ended stage. The $S11$ and $S21$ parameters of the LP-MN are plotted in Fig. 3(a), indicating excellent filter characteristics of the LP-MN to provide a good impedance matching over a wide frequency range. Following a similar impedance-matching procedure, we could obtain the design values for the input MN. Fig. 3(b) shows the overall circuit schematic of the proposed push–pull PC Class-E/F$_3$ PA.

IV. Simulation and Measurement Results

To verify the feasibility of the proposed circuit, a designed prototype is fabricated on a 1.6 mm FR-4 substrate (ε_r is 4.2, loss tangent is 0.02), as illustrated in Fig. 4(a) which size is 133 × 102 mm. Two MRF6V2300N transistors are used in the design with a 12 V drain supply voltage, while the gate voltage is biased at $V_{GS} \approx 2$ V. In the measurement, the operation frequencies are performed from 7 to 9.4 MHz, and two path square waves with a phase difference of 180° between the two gate inputs are employed to switch on-off states. Fig. 4(b) depicts the measured V_{DS}, I_{DS} drain waveforms of S_1 and S_2 at 8.2 MHz. The S_1 and S_2 exhibit opposite operating states under on-off conditions, which confirms the validation of the design procedure.

The measured drain efficiency (DE) and power gain versus P_{out} of the designed PA are shown in Fig. 5(a). A saturated P_{out} of 41.78–42.05 dBm is reached with a DE of 83.74%–89.54%. The power gain of the push–pull PC Class-E/F$_3$ PA at the saturated P_{out} is 15.95–18.44 dB. Fig. 5(b) compares the simulation and measurement, which include

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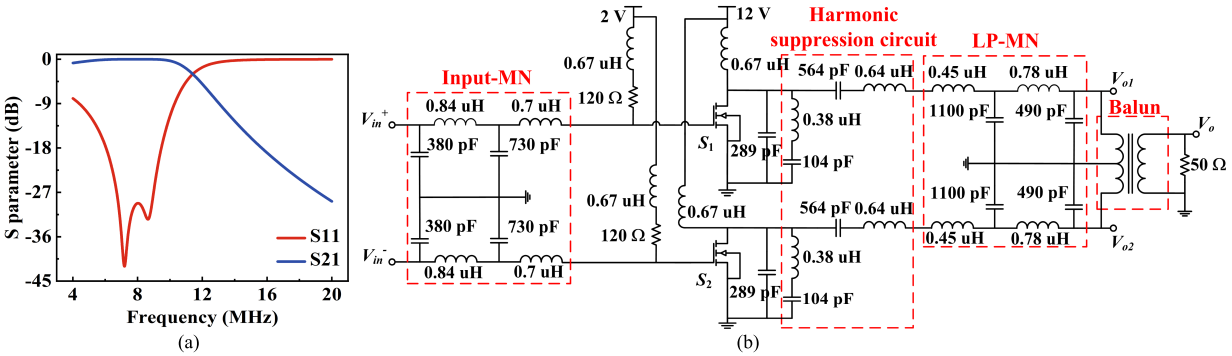


Fig. 3. (a) Simulated $S11$ and $S21$ parameters of the LP-MN. (b) Proposed push–pull PC Class-E/F$_3$ with D-RCT topology.

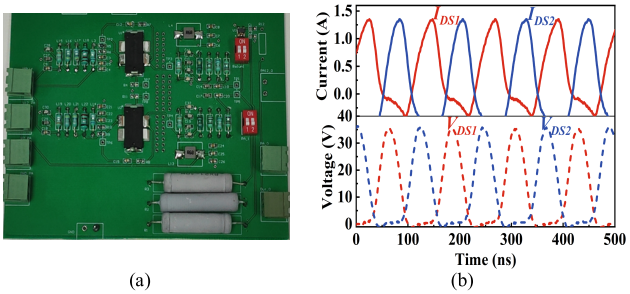


Fig. 4. (a) Fabricated PA. (b) Measured drain waveforms of S_1 and S_2 at 8.2 MHz.

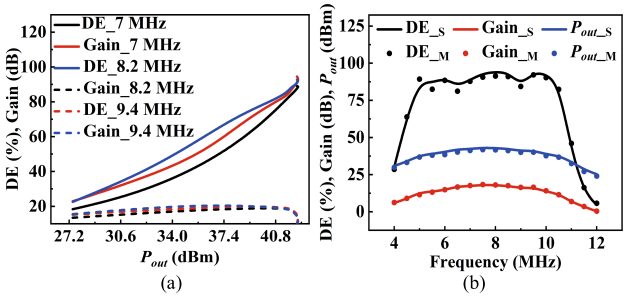


Fig. 5. (a) Measured DE and power gain versus P_{out}. (b) Simulated (S) and measured (M) DE, gain, and P_{out} versus frequency.

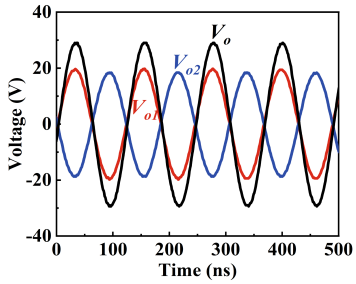


Fig. 6. Measured balanced and unbalanced output voltage waveforms at 8.2 MHz.

DE, P_{out}, and gain from 5 to 10.5 MHz. The designed PA demonstrates a DE of 85.12%–94.1% and 82.35%–91.32% in simulated and measured results, respectively. In addition, the proposed PA attains a measured P_{out} ranging from 36.83 to 41.84 dBm, along with a gain between 11.43 and 18.24 dB in the frequency range of 5–10.5 MHz. According to Fig. 5(b), the differences exist in the measurement and simulation due to the inaccuracy of the transistor model and the process error of the fabricated circuit, probably. The maximum P_{out} of the PA is 41.84 dBm (15.3 W), which is larger than 40 dBm (10 W), the difference is mainly due to the fact that the series resonator L_3–C_3 is not a short-circuit state at other operating

TABLE I
COMPARISON WITH REPORTED HIGH-FREQUENCY BROADBAND PAS

Ref	PA mode	Freq (MHz)	Gain (dB)	P_{out} (dBm)	DE (%)
[17]	Class-E	5.4-10.2	N/A	47.4	55-90
[18]	Class-S	5-15	7.61-18.35	21.13-31.87	N/A
[19]	Class-E	6.25-9	N/A	46.4-47	70-92
[20]	Class-E^{-1}	5.4-20	17.75-22.87	N/A	30-55.34
[21]	Class-E	25-31	32.4-42.5	33-42.6	75
[22]	PC Class-E	7-9.4	19.95-23.11	36.92-38.57	82.15-86.3
This work	**Push-pull PC Class-E/F$_3$**	**5-10.5**	**11.43-18.24**	**36.83-41.84**	**82.35-91.32**

frequencies, which may cause the drain voltage to increase or decrease, resulting in corresponding differences in P_{out}.

The harmonic suppression effect of the proposed push–pull structure can be observed by measuring the voltage waveforms of the balanced and unbalanced ports. The measured voltage waveforms at the unbalanced output (V_o) of the push–pull PA as well as the two balanced output (V_{o1}, V_{o2}) are shown in Fig. 6. It is noticed that the phase difference between the two differential balanced outputs deviates by 180°, and the higher harmonic suppression results in better sinusoidal signals for both the balanced and unbalanced output voltage waveforms. Compared to the single-ended scheme, the proposed topology provides enhanced efficiency and better sinusoidal waveform along with the cancellation of the even-order distortions. Meanwhile, a second-order Chebyshev LP MN prototype is used to achieve impedance transform, which will further filter the high-order harmonics and improve the quality of the output signal.

Table I shows the comparison of the proposed PA with other high-frequency PAs. By comparing with published works, the proposed PA exhibits good output performance.

V. CONCLUSION

A broadband push–pull PC Class-E/F$_3$ PA with D-RCT and harmonic suppression has been analyzed and implemented using two transistors for verification. The $3f_0$ harmonic control circuit L_3–C_3 presents a low impedance to the ground, whereas the series resonant circuit L_0–C_0 provides an open circuit to higher harmonics. Results show that the circuit presents good $2f_0$ and $3f_0$ harmonic suppression over 50 dBc in the frequency range of 6–12 MHz, reducing the total harmonic distortion of the PA. In addition, the PA can deliver a maximum DE of 91.32% and a maximum P_{out} of 41.84 dBm. Therefore, the proposed push–pull PC Class-E/F$_3$ PA can be a potential candidate in RFID and IoT applications.

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